Lecture 7 – Integration by Parts

67-272 · Fall 2026

Today: when substitution (Lecture 6) does not work because the integrand is a **product of two unrelated functions**.

> Integration by parts: $\int u\,dv = uv - \int v\,du$

It comes from the product rule: $(uv)' = u'v + uv'$. Integrate both sides and rearrange.

Choosing u: the LIATE rule

Pick u as the function that comes first in this list. Everything else is dv.

1. **L**ogarithmic – ln x
2. **I**nverse trig – arctan x
3. **A**lgebraic – x^2, 3x
4. **T**rigonometric – sin x, cos x
5. **E**xponential – e^x

Goal: du should be simpler than u, and dv must be easy to integrate.

Worked example: ∫ x eˣ dx

By LIATE, x is Algebraic and eˣ is Exponential, so $u = x$ and $dv = e^x\,dx$.

Then $du = dx$ and $v = e^x$.

$$\int x\,e^x\,dx = x\,e^x - \int e^x\,dx = x\,e^x - e^x + C$$

Check: differentiate the answer and you should get back $x\,e^x$.

Common mistake: choosing $u = e^x$. Then $v = x^2/2$ and the new integral is harder, not easier.

Practice and Problem Set 5

Try these before Thursday. Use LIATE to pick u and show each step.

1. $\int x \cos x \, dx$
2. $\int \ln x \, dx$ (hint: take dv = dx)
3. $\int x^2 e^x \, dx$ (you will use parts twice)

Problem Set 5 problems 1–4 use integration by parts. Office hours: Tue and Thu, 4:00–5:30pm.